



Deeper Dive Into Hardware Purchasing Suggestions

Tuesday 8/26



Debrief

Main goal of this meeting:

- Dive deeper into hardware
- Provide suggestions for purchasing hardware
- Discuss our next meeting
- Answer questions and receive feedback



PCIe Expansion Cards

- All connect via the rectangular “Peripheral Component Interconnect Express” slot generally near the bottom of the motherboard

- Hundreds of different kinds, namely:

GPUs

M.2 Expansion cards

USB Expansion cards

Network expansion cards (NICs)

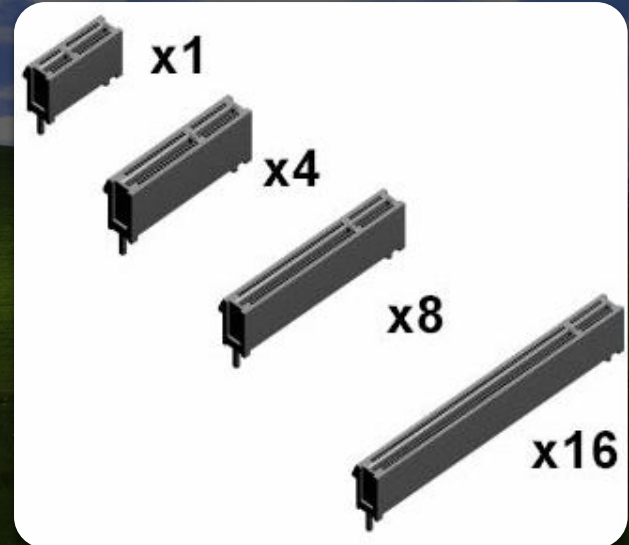
WiFi Cards

Audio Cards

Serial Cards

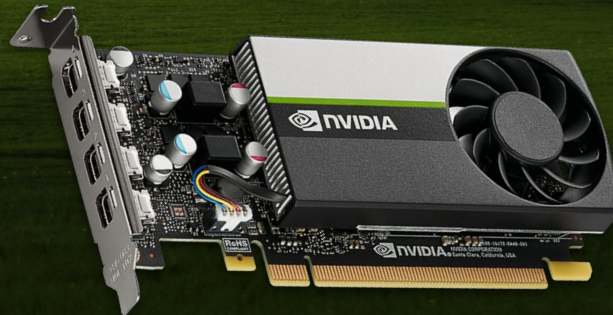
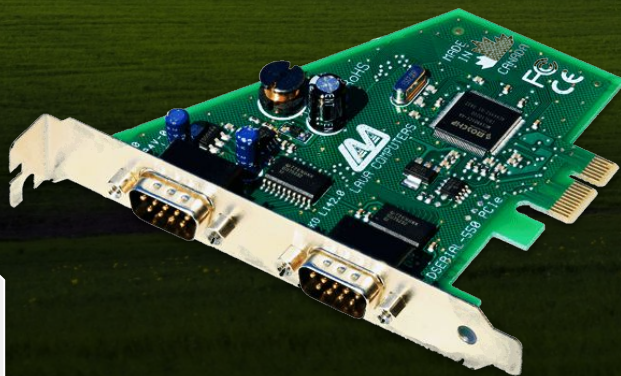
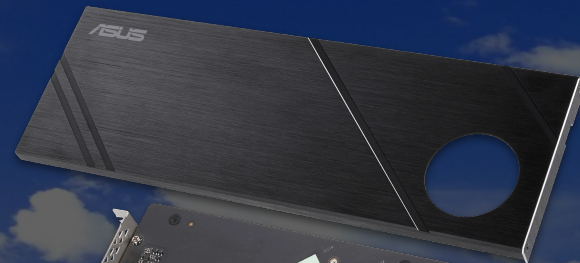
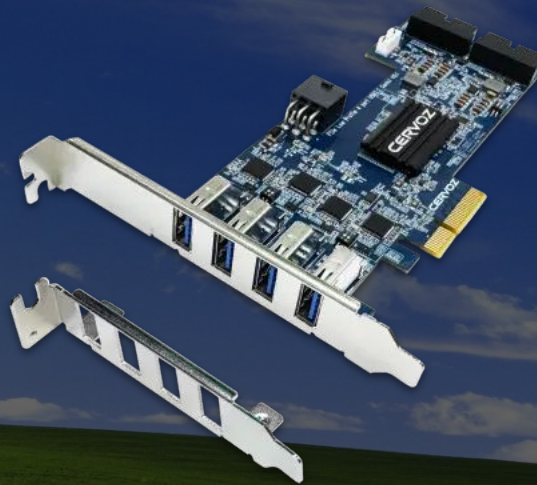
AI Acceleration Cards

Modems (PCI)



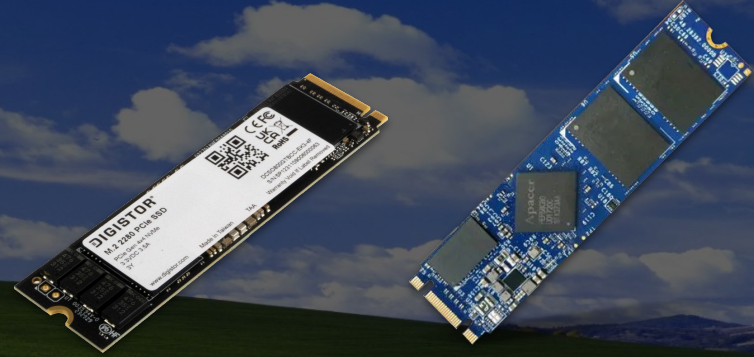
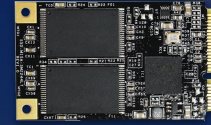


PCIe Expansion Cards Cont.



NVMe / SATA M.2

- M.2 is the connection standard for both “Non-Volatile Memory Express” and “Serial Advanced Technology Attachment”
- NVMe is significantly faster, utilizing PCIe lanes for communication versus older, slower SATA



B Key



M Key



B + M Key



Motherboard CPU Socket Types

(The cpu MUST match the motherboard socket type)

AMD

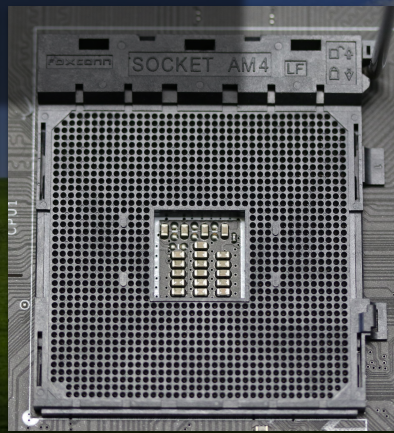
- AM4 and AM5 are the most common socket types on modern motherboards
- Newer AMD motherboards only support AM5
- Can use older AM4 motherboards, but limits upgradeability

Intel

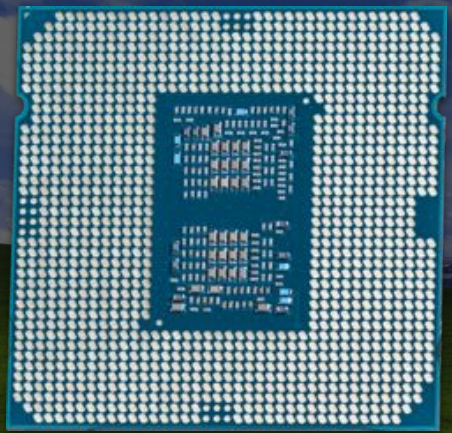
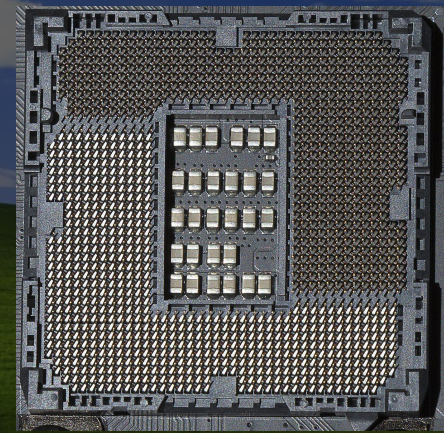
- Intel only uses LGA socket types on modern motherboards
- All post 2004 CPUs use pins on the motherboard VS the CPU
- All Intel CPU socket types must match exactly to function properly



Motherboard CPU Socket Types



AM4 vs AM5



LGA 1200



Central Processing

What should you buy?

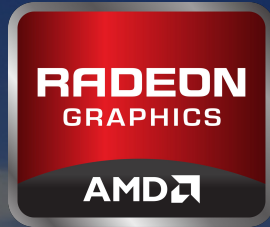


- The best brand for gaming due to the 3D V-Cache technology
- Very budget friendly due to their sockets' long lifespans
- Very power efficient
- Excellent for Productivity (photo/video editing, etc.)
- Relatively inexpensive though mobo generations do not last as long as AMD



Graphics Processing

What should you buy?



- Very budget-friendly
- Can be hard to find at MSRP
- Best use case is gaming; performance is somewhat lacking in productivity vs Nvidia
- Generous with VRAM

*VRAM is RAM for the GPU



nVIDIA

- Decently expensive
- Can be used for productivity and streaming (better than AMD)
- Very stingy with VRAM
- The highest possible consumer performer with the 5090 (AMD lacks a competitor)



Brand?

What should you buy?

- Generally most of these are up to you, but some notable brands

are:

Noctua - Coolers, Fans

Lian Li - Cases

Corsair - Cases, Coolers, PSUs

Cooler Master - Coolers, Fans

NZXT - Cases

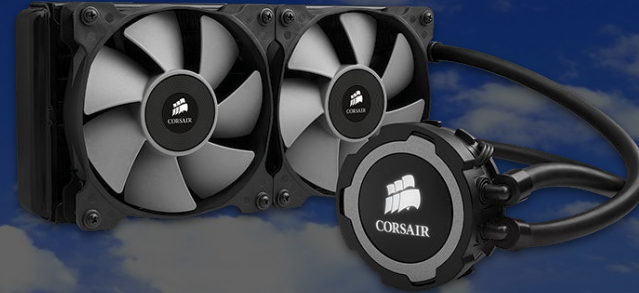
Arctic - Fans

- Staying with one brand ecosystem may make PC building a bit easier



Which CPU Cooler is best for you?

- Depends entirely on budget and what CPU you use
- For modern CPUs, small top-down coolers may not be effective enough, requiring a 'tower' or AIO cooler
- Air cooler includes simple setup, although they take up more space
- AIO includes a more complex setup, giving similar results to air cooler.





Extra Information Purchasing Knowledge

Buying used:

- Make sure you are getting what you pay for. This includes:
 - 1) Checking branding and model / model numbers
 - 2) Requesting to see the device power on and function
 - 3) Visually inspect the device for damage or voids in warranty
 - 4) Ensure you are paying an appropriate price for the device
 - 5) Ensure you aren't paying too much for scalped or resold cards

Buying new:

- Compare competitor prices and keep an eye out for sales
- If online, ensure you purchase from a reputable source, such as Amazon (shipped by and sold by Amazon preferably)
- Patience is a virtue, prices will always drop significantly



Extra Information

- Your motherboard has limited PCIe / SATA lanes, take up too many of those lanes and your devices will be slowed down significantly
- Many laptop share the exact same standards and devices as previously stated, but they are simply smaller and non-upgradable
- Fun fact: In most cases (but not as much today) building a desktop computer was cheaper than purchasing a pre-built one.



Our Mission Progress

Digital Bathroom Pass

Mission progress:

The general framework of the bathroom pass is complete, all we need now is more styling and increased security.

It should be ready to be deployed within the coming week

Feel free to speak to an officer if you have any ideas or recommendations or would like to view our progress!

[@app.route\(room/<room_number>\)](#)



Next Meeting

Tuesday 9/2

Ground plan:

- Introduce software
- Discuss the current mission





Info

Discord



Website



Instagram

